AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM354
Software Engineering

Final Examination
Spring Semester 2025/2026

Date: 4 /Jun./2026

AOU EGYPT

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 3 Hours

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **Not** allowed.

Part I – Multiple Choice Questions — 5 Marks

*Answer **all** the following multiple-choice questions. Each question is worth 1 Mark.*

1. Which of the following activities for the development of good software system is concerned with controlling the cost of development?
   a. Design b. Maintenance c. Quality management d. Project management

2. What do you call an association with a role name "/parent"?
   a. Derived association b. Class association c. Recursive association d. Qualified association

3. To which non-functional requirement would you categorize the following: "Users should be able to use the 'add to cart' link from any page listing items using a single click"
   a. Look-and-Feel b. Usability c. Performance d. Maintainability

4. Which architecture is made of similar components, where each component can request service from any other component?
   a. Call-return b. Service-Oriented c. Peer-To-Peer d. Data Flow

5. Which title is most suitable to describe the communication between the Consumer and the Registry in SOA?
   a. Find b. Bind and invoke c. Publish d. Describe

Part II – Short Answer Questions — 15 Marks

*Answer **all** the following short answer questions.*

1. Given the below code, calculate the Cyclomatic complexity for the "drive" method and the CBO (Coupling-between-objects) for the "Car" class (**IMPORTANT**: show how each value was calculated): **[5 Marks]**

```
public class Car {
    private Engine engine;
    private FuelSystem fuel;
    public Car(Engine e, FuelSystem f) {
        this.engine = e;    this.fuel = f;
    }
    public void drive(int speed, int fuelLevel, int temp) {
        if (engine.start() && fuel.hasFuel(fuelLevel)) {
            for (int i = 0; i < 2; i++) {
                if (!engine.checkTemp(temp)) {
                    System.out.println("Overheating!"); break;
                }
                switch (speed / 30) {
                    case 0: System.out.println("Slow"); break;
                    case 1: System.out.println("Moderate"); break;
                }
            }
        } else {
            System.out.println("Cannot drive");
        }
    }
}
```

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2. What approaches and methodologies are used in the development of critical systems to ensure correctness, minimize errors, and enable traceability from requirements through to implementation?

[3 Marks]

3. State 4 reasons why reuse is desirable in software engineering.

[4 Marks]

4. Compare the Client-Server architecture to the Independent Components architecture.

[3 Marks]

Part III – Problem Solving Questions 30 Marks

*Answer **all** the following problem-solving questions. Each question is worth 10 Marks.*

1. Develop an activity diagram for the process of filling up a job application online and make sure to cover the following:
   a- Must include a minimum of 8 activities
   b- Must have at least one decision with its respective merge node
   c- Must have at least 2 roles represented by swimlanes
   d- Must use the synchronization bars (Fork and Join) at least once

2. Providing a special discount on a product invokes the following method applyDiscount(employeeID, percentile)
   The employeeID should be between 1 and 500
   The percentile should be between 5 and 25
   a- Using the partitioning technique of black box testing, provide suitable test values
   b- Explain “random testing” and provide an example for the reserve method
   c- Explain “error guessing” and provide an example for the reserve method

3. Develop a sequence diagram for the process of creating events for a calendar app with the following requirements:
   a- Create 3 objects aUserInterface, app:CalendarApp, profile:OnlineProfile.
   b- Create at least 5 messages as you find suitable with at least 2 self-messages.
   c- Use the opt fragment once (Example: Event created must make sure that date has not passed).
   d- Use the loop fragment once to demonstrate how 3 events can be created.
   e- Once the event and alarm are created, a message must be sent to the profile object to update the online calendar.

End of questions

ACCEPT